

FRESNO YOSEMITE, CA, USA

WMO: 723890

Lat: 36.780N

Lon: 119.719W

Elev: 102

StdP: 100.11

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 93193

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	0.3	1.6	-5.1	2.5	14.9	-3.1	2.9	13.4	8.3	11.5	7.4	11.2	0.9	90	0.446

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.2	39.9	20.9	38.5	20.3	37.1	19.9	22.9	36.1	21.9	34.8	21.1	33.9	3.6	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.4	13.4	29.3	17.1	12.4	28.6	16.0	11.5	27.5	67.7	35.9	64.2	34.8	61.2	33.8	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
8.2	7.3	6.4	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-1.5	42.4	1.4	1.1	-2.5	43.1	-3.3	43.7	-4.1	44.3	-5.2	45.1
			-2.1	24.2	1.5	2.0	-3.2	25.7	-4.1	26.9	-4.9	28.0	-5.9	29.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.5	9.2	11.2	14.3	16.9	21.1	25.5	28.8	28.0	25.0	19.2	13.0	8.9
	DBStd	7.67	2.78	2.74	3.29	3.89	3.95	3.78	2.84	2.60	3.37	3.35	3.26	2.84
	HDD10.0	137	49	17	4	1	0	0	0	0	0	0	10	57
	HDD18.3	1188	283	199	131	73	17	1	0	0	1	29	160	294
	CDD10.0	3226	24	51	137	207	345	463	582	558	451	284	102	21
	CDD18.3	1235	0	0	5	29	103	215	324	299	202	56	2	0
	CDH23.3	16446	0	2	90	394	1323	2925	4594	4021	2452	613	31	0
	CDH26.7	9277	0	0	14	136	608	1669	2867	2443	1328	208	3	0
Wind	WSAvg	2.6	1.8	2.1	2.6	3.2	3.7	3.8	3.3	3.1	2.6	2.0	1.7	1.7
Precipitation	PrecAvg	279	54	50	50	24	9	4	0	0	4	15	32	38
	PrecMax	549	217	155	184	112	42	49	6	6	30	62	89	108
	PrecMin	154	1	0	0	0	0	0	0	0	0	0	0	0
	PrecStd	98	43	44	40	26	13	10	1	1	9	18	25	26
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.2	23.3	28.6	33.2	37.4	40.7	41.9	41.3	39.2	34.4	26.5	19.4
		MCWB	11.7	14.1	15.8	17.1	19.3	21.1	21.5	21.6	20.1	18.6	15.3	12.8
	2%	DB	17.8	20.6	25.8	30.3	34.8	38.6	40.1	39.3	37.2	31.1	23.8	17.2
		MCWB	12.1	13.8	15.5	16.5	18.6	20.2	21.1	20.7	19.6	17.4	14.6	11.8
	5%	DB	15.9	18.6	23.7	27.9	32.5	36.7	38.7	37.8	35.6	28.9	21.5	15.5
		MCWB	11.5	12.6	14.6	15.7	17.6	19.4	20.6	20.0	19.2	16.5	13.9	11.1
	10%	DB	14.3	16.8	21.5	25.2	30.1	34.7	37.1	36.3	33.7	26.9	19.3	13.9
		MCWB	11.0	12.0	13.9	14.9	16.8	18.8	20.2	19.7	18.7	15.8	13.0	10.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.0	15.8	18.0	19.2	21.1	23.4	24.1	24.7	22.4	20.0	16.9	15.0
		MCDB	17.3	19.9	25.3	28.3	33.6	36.8	38.0	37.0	34.5	30.1	23.5	17.6
	2%	WB	13.3	14.6	16.7	17.5	19.8	22.1	22.9	22.5	21.2	18.8	15.7	13.3
		MCDB	15.6	19.0	23.8	27.4	32.4	35.2	36.8	34.8	33.1	28.5	21.7	15.3
	5%	WB	12.3	13.6	15.5	16.5	18.6	20.8	21.9	21.4	20.2	17.6	14.7	12.2
		MCDB	14.6	17.6	21.9	26.0	30.3	33.3	35.2	34.2	32.5	27.0	20.0	14.3
	10%	WB	11.2	12.5	14.4	15.4	17.5	19.8	21.1	20.5	19.4	16.6	13.7	11.0
		MCDB	13.7	16.0	20.2	24.1	28.4	32.3	34.2	33.5	31.8	25.2	18.2	13.2
Mean Daily Temperature Range	5% DB	MDBR	8.8	10.1	11.8	13.0	14.2	15.8	16.2	16.3	15.4	13.6	11.0	8.9
		MCDBR	11.8	12.9	15.2	16.7	17.0	17.2	16.7	17.1	17.1	16.3	14.4	11.3
		MCWBR	6.9	6.8	6.4	5.9	5.5	5.1	5.0	4.9	5.1	5.8	6.7	6.6
	5% WB	MCDBR	8.8	11.2	13.7	15.2	16.1	16.3	15.7	16.2	15.8	14.7	12.4	7.6
		MCWBR	5.6	6.3	6.4	5.8	5.6	5.3	5.4	5.3	5.1	5.4	6.3	5.0
Clear-Sky Solar Irradiance	taub	0.359	0.372	0.395	0.378	0.368	0.364	0.383	0.368	0.371	0.390	0.380	0.371	
	taud	2.324	2.304	2.239	2.331	2.387	2.399	2.374	2.389	2.391	2.283	2.253	2.266	
	Ebn at Noon	815	853	867	903	914	914	893	901	881	820	778	767	
	Edh at Noon	97	111	130	125	120	118	121	116	110	112	103	96	
All-Sky Solar Radiation	RadAvg	2.44	3.48	4.96	6.31	7.50	8.41	8.10	7.43	6.17	4.48	3.03	2.22	
	RadStd	0.35	0.41	0.43	0.48	0.43	0.31	0.30	0.23	0.19	0.26	0.17	0.31	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.56	N/A	-1.22	N/A	N/A	-0.80	-38	-124	+191	+114
(47)	Regional (23 neighbors)	+0.51	N/A	-1.14	+0.44	N/A	N/A	N/A	-136	+167	+53

Nomenclature: See separate page